25901869 **Reg.No: ……**

Name: ……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION APRIL - 2026

Second SEMESTER

Skill Development Components (SDC) – MG2SDCREM101 – SOLAR PHOTOVOLTAIC SYSTEMS

(2025 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **15** questions

Each question carries **1** mark

1. Write the standard unit for PV module rating. [A]/[CO1]

Answer: Watt-peak (Wp)

2. Write one example of second generation solar cell. [A]/[CO1]

Answer: Thin-film solar cell (e.g., CdTe or CIGS)

3. Write the definition of a solar PV module. [A]/[CO1]

Answer: A PV module is an assembly of interconnected solar cells encapsulated to generate electrical power from sunlight.

4. Write the function of diode in the equivalent circuit. [A]/[CO1]

Answer: It represents the p-n junction behavior and controls current flow in one direction.

5. Associate shading with reduction in PV module power output. [U]/[CO1.CO2]

Answer: Shading reduces current generation, leading to significant power loss.

6. Relate load demand to battery capacity in a standalone PV system. [U]/[CO2]

Answer: Higher load demand requires higher battery capacity for sufficient energy storage.

7. Write one purpose of a multi-branch connector. [A]/[CO1]

Answer: To connect multiple PV strings in parallel.

8. Classify the effect of shading on a series-connected PV string. [U]/[CO2]

Answer: It reduces overall current and may cause hotspot issues.

9. Show the symbol for a series connection. [A]/[CO1]

Answer: Components connected end-to-end (+ to -).

10. Select one method to ensure stable operation of a grid-connected PV system. [A]/[CO3]

Answer: Maximum Power Point Tracking (MPPT)

11. Define overcurrent protection in the context of PV Combiner Boxes. [K]/[CO4]

Answer: Protection using fuses or breakers to prevent excessive current damage.

12. Recognize the effect of loose connections inside a junction box on system performance[K]/[CO4]

Answer: Causes power loss, overheating, and possible failure.

13. Explain why insulation resistance decreases with moisture (in one sentence). [U]/[CO4]

Answer: Moisture increases conductivity, reducing insulation resistance.

14. Select the device used for isolating the AC supply in ACDB. [K]/[CO4]

Answer: AC Isolator or Circuit Breaker

15. Relate temperature rise to reduction in wire current carrying capacity. [U]/[CO4]

Answer: Higher temperature reduces current carrying capacity due to increased resistance.

16. Write one electrical factor affecting PV module efficiency. [A]/[CO1]

Answer: Temperature

17. Relate module voltage rating to battery voltage in a standalone system. [U]/[CO2,CO5]

Answer: Module voltage must be higher than battery voltage for proper charging.

[1X15=15]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

18. Illustrate the contrast between maximum power point and open circuit condition using the I–V characteristics of a solar cell. [A][CO1]

Answer:
The performance of a solar cell is best understood using its I–V (current-voltage) characteristics curve, which shows the relationship between output current and voltage under given conditions.

In the open circuit condition, the terminals of the solar cell are not connected to any load. As a result, the current becomes zero, while the voltage reaches its maximum value known as the open circuit voltage (Voc). At this point, although voltage is maximum, no power is delivered since current is zero.

In contrast, the maximum power point (MPP) is the point on the I–V curve where the product of current and voltage is maximum. This point lies between the open circuit voltage (Voc) and short circuit current (Isc). At MPP, the solar cell operates at its highest efficiency and delivers maximum usable power.

In practical PV systems, Maximum Power Point Tracking (MPPT) techniques are used to ensure that the system always operates at this optimal point despite changing environmental conditions.

Thus, while the open circuit condition represents maximum voltage with zero power output, the maximum power point represents the most efficient operating condition with maximum power delivery.

19. Present the comparison between the function of a bypass diode and the condition when no bypass diode is used in a PV module. [A][CO1]

Answer:
Bypass diodes are essential components used in photovoltaic modules to enhance performance and protect the system under partial shading conditions.

A bypass diode is connected in parallel with a group of solar cells. When some cells are shaded or damaged, the diode provides an alternate path for current to flow, thereby preventing reverse biasing of those cells. This helps to avoid the formation of hotspots, which can damage the module. It also ensures that the overall power output is not drastically reduced.

In the absence of a bypass diode, shaded cells behave like resistive loads and start dissipating power instead of generating it. This leads to overheating, hotspot formation, and possible permanent damage to the solar cells. Additionally, the overall output power of the module drops significantly.

Therefore, bypass diodes improve the reliability, safety, and efficiency of PV modules, especially under non-uniform illumination conditions.

20. Write the advantages and limitations of lithium-ion batteries compared to lead-acid batteries for solar applications. [A][CO3]

Answer:
Battery storage plays a crucial role in solar power systems, and the choice of battery type significantly affects performance and efficiency.

Lithium-ion batteries offer several advantages over traditional lead-acid batteries. They have a higher energy density, meaning they can store more energy in a smaller size. They also have a longer lifespan with more charge-discharge cycles, higher efficiency (typically 90–95%), and require minimal maintenance. Additionally, they are lightweight and compact.

However, lithium-ion batteries have certain limitations. They have a higher initial cost and require a Battery Management System (BMS) for safe operation. They are also sensitive to high temperatures and need proper control systems.

On the other hand, lead-acid batteries are cheaper and widely available but are heavier, have lower efficiency, shorter lifespan, and require regular maintenance.

Thus, lithium-ion batteries are more suitable for modern solar applications due to their superior performance, despite their higher cost.

21. Explain the factors affecting voltage drop in DC cabling. [U][CO4]

Answer:
Voltage drop in DC cabling is an important factor that affects the efficiency of photovoltaic systems. It occurs due to the resistance offered by the conductors as current flows through them.

Several factors influence voltage drop. The length of the cable plays a major role, as longer cables result in higher resistance and greater voltage drop. The amount of current flowing through the conductor also affects the drop, with higher current leading to increased losses. The cross-sectional area of the conductor is another key factor; smaller conductors have higher resistance and hence more voltage drop.

The material of the conductor also matters, with copper having lower resistance compared to aluminum. Additionally, temperature affects resistance, as higher temperatures increase resistance and thereby increase voltage drop.

To minimize voltage drop, proper cable sizing, shorter cable lengths, and high-quality conductive materials should be used.

Thus, careful design of DC cabling ensures improved efficiency and reduced power losses in PV systems.

22. Classify the electrical and physical parameters of a PV module and explain their importance in system design. [A][CO1]

Answer:
The parameters of a photovoltaic module are broadly classified into electrical and physical parameters, both of which are important for system design.

Electrical parameters include open circuit voltage (Voc), short circuit current (Isc), maximum power (Pmax), and voltage and current at maximum power point (Vmpp and Impp). These parameters are crucial for selecting compatible inverters, charge controllers, and for proper system sizing.

Physical parameters include the size, dimensions, weight, and mounting type of the module. These factors influence installation, structural support design, and space requirements.

Understanding both electrical and physical parameters ensures that the PV system operates efficiently and safely. Proper selection and integration of modules help achieve optimal performance and long-term reliability.

Hence, these parameters play a vital role in the design and implementation of solar PV systems

[5X3=15]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

23. Present the differences between field measurement and STC measurement of PV module parameters, including the effect on efficiency and output. [A][CO1]

    Answer:
    The performance of photovoltaic (PV) modules is evaluated under two main conditions: Standard Test Conditions (STC) and field conditions.

Standard Test Conditions (STC) are controlled laboratory conditions defined as solar irradiance of 1000 W/m², cell temperature of 25°C, and air mass of 1.5. These conditions provide a uniform basis for comparing different PV modules and are used to specify the rated power of a module. Under STC, the output values such as voltage, current, and efficiency are ideal and consistent.

In contrast, field measurements are taken under actual environmental conditions where parameters such as solar irradiance, ambient temperature, wind speed, dust, and shading vary continuously.

Because of these variations, the actual performance of PV modules in the field differs significantly from STC values.

One major factor affecting performance is temperature. As temperature increases, the open circuit voltage (Voc) decreases, leading to a reduction in power output. Similarly, variations in irradiance directly affect the current output of the module. Dust accumulation and partial shading also reduce the effective irradiance reaching the panel surface.

As a result, the efficiency and output power in field conditions are usually lower than STC ratings. Therefore, while STC provides a standard reference for comparison and rating, field measurements reflect the real performance of the PV system.

Thus, understanding both STC and field conditions is essential for accurate system design, performance prediction, and efficiency evaluation.

24. Discuss the impact of light intensity and temperature on the output characteristics of a silicon solar cell. [U][CO2]

    Answer:
    The output characteristics of a silicon solar cell are significantly influenced by environmental factors, particularly light intensity (irradiance) and temperature. These factors directly affect the current, voltage, and overall efficiency of the solar cell.

When light intensity increases, more photons strike the surface of the solar cell, resulting in the generation of a greater number of electron-hole pairs. This increases the short circuit current (Isc) almost linearly with irradiance. As a result, the overall power output of the solar cell also increases. However, the effect of irradiance on voltage is relatively small compared to its effect on current.

On the other hand, temperature has an inverse effect on performance. As temperature increases, the semiconductor properties of the solar cell change, leading to increased recombination of charge carriers. This results in a reduction in the open circuit voltage (Voc). Since power output is a product of voltage and current, the decrease in voltage leads to a reduction in overall efficiency.

In real conditions, high solar irradiance is often accompanied by high temperature. While increased irradiance tends to increase power output, the simultaneous rise in temperature reduces efficiency. Therefore, the net effect depends on the balance between these two factors.

To improve performance, proper system design should include adequate ventilation and cooling mechanisms to reduce temperature rise.

Thus, while higher light intensity enhances power generation, increased temperature negatively affects efficiency, making thermal management crucial in PV systems.

25. Illustrate the proper charging and discharging procedure to maximize battery life. [A][CO3]

Answer:
Battery life in solar energy systems is greatly influenced by proper charging and discharging practices. Efficient battery management ensures long lifespan, reliable performance, and reduced maintenance costs.

During the charging process, it is important to use a suitable charge controller such as PWM (Pulse Width Modulation) or MPPT (Maximum Power Point Tracking). These controllers regulate the voltage and current supplied to the battery, preventing overcharging. Overcharging can lead to excessive

heat generation, electrolyte loss (in lead-acid batteries), and internal damage. Charging should be done in stages such as bulk, absorption, and float to ensure complete and safe charging.

During the discharging process, deep discharge should be avoided as it significantly reduces battery life. Each battery has a recommended Depth of Discharge (DoD), and exceeding this limit can cause permanent damage. Maintaining a moderate discharge level improves cycle life and efficiency.

Temperature also plays an important role. Batteries should be operated within the recommended temperature range, as high temperatures accelerate chemical degradation, while very low temperatures reduce capacity.

Regular maintenance, including checking electrolyte levels (for lead-acid batteries), cleaning terminals, and monitoring voltage levels, is essential for optimal performance.

Thus, by following proper charging stages, avoiding deep discharge, and maintaining suitable operating conditions, the lifespan and efficiency of batteries can be significantly improved.

26. Explain the procedure for designing an effective grounding system for a substation or industrial installation. [U][CO4]

Answer:
An effective grounding system is essential for ensuring the safety of personnel and equipment in substations and industrial installations. It provides a low-resistance path for fault currents and helps maintain system stability.

The design process begins with soil resistivity measurement, which determines the ability of the ground to conduct electrical current. This is typically done using methods such as the Wenner four-pin method. The results help in selecting the type and depth of grounding electrodes.

Next, a grounding grid (earth mat) is designed. This consists of a network of conductors buried in the ground, arranged in a grid pattern. The grid ensures uniform distribution of fault current and reduces potential differences across the surface.

Grounding electrodes, such as rods, plates, or pipes, are then installed deep into the soil to achieve low resistance. These electrodes are connected to the grid to enhance conductivity.

All electrical equipment, structures, and metallic parts are properly bonded to the grounding system to ensure that they remain at the same potential, thereby preventing electric shock hazards.

The system is then tested to ensure that the ground resistance is within safe limits, typically less than 1 ohm for substations. Additionally, the design must consider step voltage and touch voltage, ensuring they remain within safe limits to protect personnel.

Proper grounding also helps in lightning protection and minimizes damage due to electrical faults.

Thus, a well-designed grounding system ensures safe dissipation of fault currents, protects equipment, and provides reliable and stable system operation.

[10X2=20]